Practical No.05

Q.1] Write code to display a message when a button is clicked.

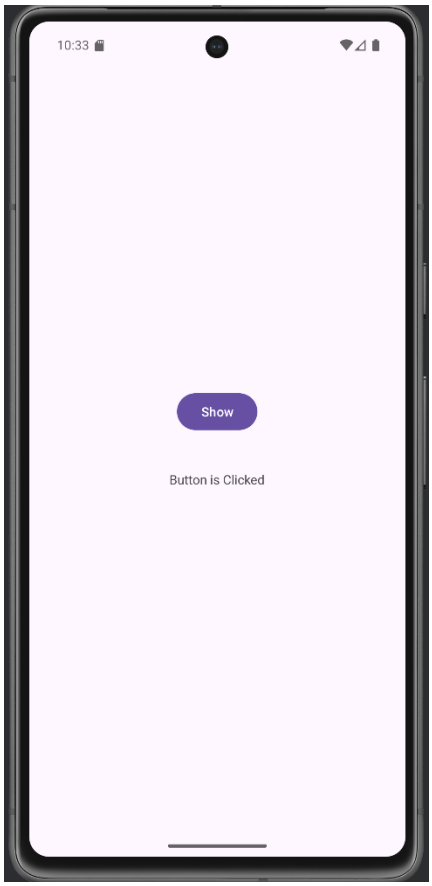
XML code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:orientation="vertical"
   android:gravity="center"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <Button
      android:id="@+id/button"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_margin="20dp"
      android:text="Show"
      />
   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_margin="20dp"
      android:text="Hello DYP"
      />
</LinearLayout>
```

Java code:

```
package com.techtitans.madpracticals;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        TextView txt = findViewById(R.id.textView);
        Button btn = findViewById(R.id.button);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                txt.setText("Button is Clicked");
            }
        });
    }
}
```

output :-



Q.2] Create a layout with all three buttons: Button,ImageButton and ToggleButtton.

XML code :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:id="@+id/main"
 android:orientation="vertical"
 android:gravity="center"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">
```

```
 <Button
 android:id="@+id/button"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_margin="20dp"
 android:text="Click"
 />
 <ImageButton
 android:id="@+id/imageButton"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_margin="20dp"
 app:srcCompat="@drawable/img_1" />
 <ToggleButton
 android:id="@+id/toggleButton"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_margin="20dp"
 android:text="ToggleButton" />
</LinearLayout>
```

Java code:

```
package com.techtitans.madpracticals;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main); }}
```

Output :-

Q.3] Change TextView Based On Toggle.

XML code :-

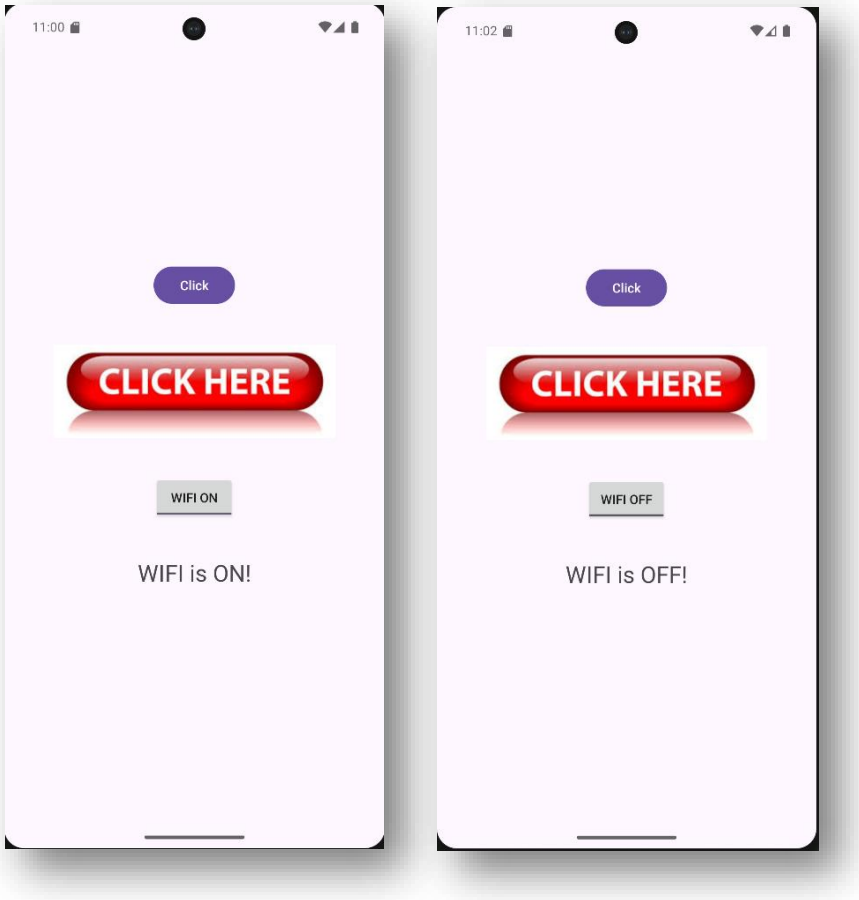
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <Button
```

```
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="Click" />

    <ImageButton
        android:id="@+id/imageButton"
        android:layout_width="304dp"
        android:layout_height="101dp"
        android:layout_margin="20dp"
        app:srcCompat="@drawable/img_1" />

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="ToggleButton"
        android:textOff="WIFI OFF"
        android:textOn="WIFI ON" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:textSize="25sp"
        android:text="Hello" />

</LinearLayout>
```

Java code :-

```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.view.View;
import android.widget.TextView;
import android.widget.ToggleButton;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        TextView txt = findViewById(R.id.textView2);
        ToggleButton tg = findViewById(R.id.toggleButton);
        tg.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                if (tg.isChecked()) {
                    txt.setText("WIFI is ON!");
                } else {
                    txt.setText("WIFI is OFF!");
                }
            }
        });
    }
}
```

output :-



Q.4] Image Button Interaction

I. Add an Image Button using an image from the draw able folder
II. On Click, display a Toast message: “Image Button Pressed.”

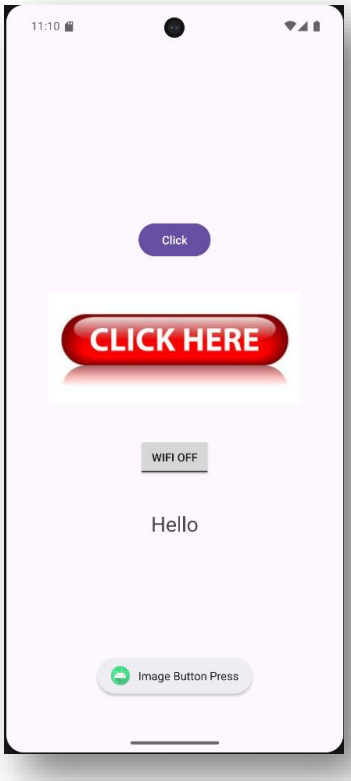
XML code :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="Click" />

    <ImageButton
        android:id="@+id/imageButton"
        android:layout_width="309dp"
        android:layout_height="137dp"
        android:layout_margin="20dp"
        app:srcCompat="@drawable/img_1" />

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:textOff="WIFI OFF"
        android:textOn="WIFI ON"
        android:text="ToggleButton"/>
    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:textSize="25sp"
        android:text="Hello" />

</LinearLayout>
```

Java code:

```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.view.View;
import android.widget.ImageButton;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        TextView txt = findViewById(R.id.textView2);
        ImageButton img = findViewById(R.id.imageButton);
        img.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Toast.makeText(getApplicationContext(), "Image Button Press",
                        Toast.LENGTH_LONG).show();

                }
        });
        ToggleButton tg = findViewById(R.id.toggleButton);
        tg.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                if (tg.isChecked()) {
                    txt.setText("WIFI is ON!");
                } else {
                    txt.setText("WIFI is OFF!");
```

```
            }
        }
    });
  }
}
```

Output :